In this paper, all excercises in Hoffman / Kunze will be solved in Complete form.

Sec 6.2.

6.2.1. Denote f_i is the Characteristic Polynomial of the given Matrix A_i.

$f_1(t)=t(t-1)$, $f_2(t)=(2-t)(1-t)+3=t^2-3t+5$, $f_3(t)=(1-t)^2-1=t^2-2t$.

In $\mathbb{R}$, f_2 ~~and~~ $\cancel{f_3}$ do not split over the given field, actually there are No solution in $\mathbb{R}$ of f_2 ~~and~~ $\cancel{f_3}$. So A_2 and $\cancel{A_3}$, ~~has no eigenvector.~~

But, $\begin{bmatrix}1&0\\0&0\end{bmatrix}\begin{bmatrix}0\\1\end{bmatrix}=0$, so $E_0=\ker A=\mathrm{span}_{\mathbb{R}}\{\binom{0}{1}\}$

And $\begin{bmatrix}1&0\\0&0\end{bmatrix}\begin{bmatrix}1\\0\end{bmatrix}=\begin{bmatrix}1\\0\end{bmatrix}$, so $E_1=\mathrm{span}_{\mathbb{R}}\{\binom{1}{0}\}$,

$$\frac{2\pm\sqrt{4-8}}{2}$$

In $\mathbb{C}^2$, A_1 has same eigenvectors. But:

$$t=\frac{3\pm\sqrt{9-20}}{2}=\frac{3\pm i\cdot\sqrt{11}}{2}$$ are roots of f_2, and

$$t=\frac{2\pm\sqrt{4-8}}{2}=1\pm i$$ are roots of f_3. So,

$$A_2-\underbrace{\left(\frac{3\pm i\sqrt{11}}{2}\right)}_{\lambda_2}\cdot I=\begin{bmatrix}\frac{1}{2}\pm i\cdot\frac{\sqrt{11}}{2} & 3\\ -1 & -\frac{1}{2}\pm i\cdot\frac{\sqrt{11}}{2}\end{bmatrix}$$

Then, $$(A_2-\lambda_2)I\cdot\binom{x}{y}=\begin{pmatrix}\left(\frac{1}{2}+i\cdot\frac{\sqrt{11}}{2}\right)x+3y\\ -x+\left(-\frac{1}{2}+i\cdot\frac{\sqrt{11}}{2}\right)y\end{pmatrix}=\binom{0}{0}$$

$$\Leftrightarrow\begin{cases}\left(\frac{1}{2}+i\cdot\frac{\sqrt{11}}{2}\right)x+3y=0\\ \left(\frac{1}{2}+i\cdot\frac{\sqrt{11}}{2}\right)x+3y=0.\end{cases}$$ (multiplying $\left(\frac{1}{2}-i\cdot\frac{\sqrt{11}}{2}\right)$ to the bottom)

So, the eigenvectors are:

$$\begin{pmatrix}-1\\ \frac{1}{3}\left(\frac{1}{2}\pm i\cdot\frac{\sqrt{11}}{2}\right)\end{pmatrix},$$ corresponding to $\frac{1}{2}\pm i\cdot\frac{\sqrt{11}}{2}$

#6.2.6. $A = \begin{bmatrix} 0 & 0 & 0 & 0 \\ a & 0 & 0 & 0 \\ 0 & b & 0 & 0 \\ 0 & 0 & c & 0 \end{bmatrix}$ over $\mathbb{R}$.

$\det(A - tI) = t^4$, and A has the only eigenvalue as 0. So,

A is diagonalizable iff $\dim \ker A = 4$.

But, the only possibility for $\dim \ker A = 4$ is $a = b = c = 0$,
because if at least one of a or b or c is not zero, then $\operatorname{rank} A \geq 1$.

#6.2.7. If $T: F^n \to F^n$ has n distinct eigenvalues, then T is diagonalizable.
proof Let $\alpha_1, \cdots, \alpha_n \in F^n$ be the eigenvectors corresponding to distinct eigenvalues.
Then, $\{\alpha_1, \cdots, \alpha_n\}$ are linearly independent, because: Suppose that

$$c_1\alpha_1 + c_2\alpha_2 + \cdots + c_n\alpha_n = 0 \quad \text{where } c_i \in F.$$

Then for any F-polynomial $f(t) \in F[t]$,

$$0 = f(T)(c_1\alpha_1 + \cdots + c_n\alpha_n) = f(c_1)\cdot\alpha_1 + \cdots + f(c_n)\alpha_n \cdots (*)$$

Take $f_i(t) = \prod_{j \neq i}(t - c_j) \Big/ \prod_{j \neq i}(c_i - c_j)$

Then, $f_i(c_j) = \delta_{ij}$, so, we can conclude $c_1 = c_2 = \cdots = c_n = 0$,

by substituting $f_1, \cdots, f_n$ to $(*)$.

#6.2.8. Let $A, B \in M_{n\times n}(F)$. If $I-AB$ is invertible, then $I-BA$ is invertible with

$$(I-BA)^{-1} = I + B\cdot(I-AB)^{-1}\cdot A$$

Proof. Let $C=(I-AB)^{-1}$. Then,

$C-ABC = C-CAB = I$, so $ABC = CAB$ (That is, C and AB commute.)

Now $(I-BA)\cdot(I+BCA) = I+BCA-BA-BABCA$

$$\begin{aligned} &= I+BCA-BA-B(C-I)A \\ &= I+BCA-BA-BCA+BA \\ &= I. \end{aligned}$$

So, proved.

#6.2.9. Using the fact above, deduce that

AB and BA have precisely the same eigenvalue in F.

Proof. If $t=0$, then

t is an eigenvalue for $AB \iff \det AB=0 \iff \det BA=0 \iff t$ 〃 BA.

If $t\neq 0$, then $\det(AB-tI)=0 \iff t^{-n}\cdot\det((AB)\cdot t^{-1}-I)=0$

$$\begin{aligned} &\iff \det((BA)\cdot t^{-1}-I)=0 \\ &\iff \det(BA-tI)=0. \end{aligned}$$

So, proved.

#6.2.10. 2 by 2 Symmetric Matrix is diagonalizable.
Proof. Let $A = \begin{pmatrix} a & c \\ c & b \end{pmatrix}$ be given. Then,

$$\det(A - tI) = (t-a)(t-b) - c^2 = t^2 - (a+b)t + ab - c^2$$

If $c=0$, then it is clearly diagonal. And, if $c \neq 0$, then $c^2 > 0$, so the charateristic polynomial has two distinct roots, denote r_1 and r_2,

$$\dim \ker(A - r_1 I) \geq 1 \quad \text{and} \quad \dim \ker(A - r_2 I) \geq 1,$$

So $\dim \ker A - r_1 I = \dim \ker A - r_2 I = 1$, so diagonalizable.

#6.2.11. If $N \in M_{2\times 2}(\mathbb{C})$ satisfies $N^2 = 0$, then either $N = 0$ or $N \sim \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$
Proof. If $N=0$, then we are done. Suppose that $N \neq 0$, but $N^2 = 0$.
(Directly, the minimal polynomial of N is t^2, so it can be triangular.
But in this, we prove more specifically.)
Since $N^2 = 0$ implies $N \cdot Nx = 0$ for all $x \in \mathbb{C}^2$, so

$$\operatorname{Im} N \subseteq \ker N.$$

Thus $\ker N \neq \{0\}$, take $\alpha_1 \in \ker N$ be non-zero.
And, since $N \neq 0$, $\ker N \neq \mathbb{C}^2$. So, take $\alpha_2 \in \mathbb{C} \setminus \ker N$ s.t $N\alpha_2 \in \ker N$.
Now, $N\alpha_1 = 0$ and $N\alpha_2 = c \cdot \alpha_1$ for some non-zero scalar $c \in F$, so

$$[L_N]_{\{\alpha_1, \alpha_2\}} = \begin{bmatrix} 0 & c \\ 0 & 0 \end{bmatrix} \quad \text{or} \quad [L_N]_{\{\alpha_2, \alpha_1\}} = \begin{bmatrix} 0 & 0 \\ c & 0 \end{bmatrix},$$

(To conclude the conversation completely, take $\alpha_2' = c^{-1}\alpha_2$).

#6.2.12. Every complex 2 by 2 Matrix is similar to $\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$ or $\begin{bmatrix} a & 0 \\ 1 & a \end{bmatrix}$.
Proof. The charatoristic polynomial of the given Matrix is must be degree 2 complex polynomial,
So, there are only two possibilites; the polynomial has only one root of multiplicity 2
or distinct two roots.
In the second case, it is diagonalizable. In the first case, let $c \in F$ be the root.
Then, $(N - cI)^2 = 0$, because $\det(N - tI) = (t-c)^2$, so $\ker N - cI \neq 0$.
So $N - cI \sim \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$, or $N \sim \begin{bmatrix} a & 0 \\ 1 & a \end{bmatrix}$.

#6.2.13.

Let $V = \{f : \mathbb{R} \to \mathbb{R} \mid f \text{ is continuous}\}$, and define

$$T : V \to V ; f \mapsto \int_0^x f(t)\, dt.$$

Then T has no eigenvalues.

Proof. Since the integral has linearity, T is a linear map.

Suppose that T has an eigenvalue, $c \in \mathbb{R}$. That is,

$$T(f) = c \cdot f(x) = \int_0^x f(t)\, dt \quad \text{for some non-zero } f \in V.$$

If $c = 0$, then $\int_0^x f(t)\, dt = 0$ for all x, so f is zero, contradiction.

If $c \neq 0$, then $c \cdot f(0) = \int_0^0 f = 0$. Moreover,

$$c \cdot F'(x) = F(x) \Rightarrow F(x) = c^{-1} e^{cx} \to f(x) = e^{cx}$$

But $e^{c0} = e^0 = 1 \neq 0$, so contradiction.

#6.2.14.

Let $A \in M_{n\times n}(F)$ be a matrix with ch. polynomial

$$(x - c_1)^{d_1} \cdots (x - c_k)^{d_k}$$

Let $V = \operatorname{span}_F \{B \in M_{n\times n} \mid AB = BA\}$, then $\dim V = d_1^2 + \cdots + d_k^2$.

Proof.

#6.3.2.

Let $a,b,c \in F$. Then,

$A = \begin{bmatrix} 0 & 0 & -c \\ 1 & 0 & -b \\ 0 & 1 & -a \end{bmatrix}$ has characteristic polynomial as x^3+ax^2+bx+c, and it is minimal polynomial for A.

Proof. $\det(xI-A) = \begin{vmatrix} x & 0 & c \\ -1 & x & b \\ 0 & -1 & x+a \end{vmatrix} = x\cdot(x(x+a)+b)+c = x^3+ax^2+bx+c$.

And,

$A^2 = \begin{bmatrix} 0 & -c & ac \\ 0 & -b & -c+ab \\ 1 & -a & -b+a^2 \end{bmatrix}$ and $A^3 = \begin{bmatrix} -c & ac & bc-a^2c \\ -b & -c+ab & ac+b^2-a^2b \\ -a & -b+a^2 & -c+2ab-a^3 \end{bmatrix}$

So, by the direct Calculation, $A^3+aA^2+bA+cI = O$.

Now, $\{I, A, A^2\}$ is linearly independent, so there is no polynomial $p(x)$ of degree 2 s.t.

$$P(A)=0$$

So, the polynomial x^3+ax^2+bx+c is minimal polynomial for A.

#6.3.3. Put $A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ -1 & -1 & 0 & 0 \\ -2 & -2 & 2 & 1 \\ 1 & 1 & -1 & 0 \end{bmatrix}$. Then, $\det(xI-A) =$

#6.3.5. Let V be an n-dimensional Vector Space and $T: V \to V$ be a linear operator. If $T^k = 0$ for some $k \in \mathbb{N}$, then $T^n = 0$.

Proof. If $k \le n$, then clear, $T^n = T^{n-k} \circ T^k = 0$.
Suppose that $k > n$. By the Cayley-Hamilton Theorem, the minimal polynomial $p(x)$ of T satisfies $\deg p(x) \le n$, since $p(x)$ divides the charictorstic polynomial of T.
So, the minimal polynomial of T also divides T^k, so $T^r = 0$ for some $r \le n$.
Now, $T^n = T^{n-r} \circ T^r = 0$.

#6.3.6. Find a 3 by 3 Matrix for which the minimal polynomial is x^2.

Proof. A has minimal polynomial as x^2 if and only if:

$$A^2 = 0 \text{ and } \{A, I\} \text{ is linearly independent.}$$

So, $A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

#6.3.7. Let $D: P_n(\mathbb{R}) \to P_n(\mathbb{R})$ be the differentiation operator. Evaluate the minimal polynomial for D.

Proof. For any polynomial $a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 \in P_n(\mathbb{R})$,

$$D^n\left(\sum a_i x^i\right) = n! \cdot a_n$$

And, so $D^{n+1} = 0$, the zero map. Now, suppose that

$$c_n D^n + c_{n-1} D^{n-1} + \cdots + c_1 D + c_0 I = 0, \quad \text{where } c_0, \cdots, c_n \in \mathbb{R}.$$

Substracting Polynomials;

$$f_0(x) = 1,\ f_1(x) = x,\ f_2(x) = \frac{1}{2!} \cdot x^2,\ f_3(x) = \frac{1}{3!} \cdot x^3,\ \cdots,\ f_n(x) = \frac{1}{n!} \cdot x^n$$

Then, the all coefficients are zero, so $\{I, D, \cdots, D^n\}$ is linearly independent.
In other say, the polynomial of degree less than $n+1$ connot be the minimal polynomial.

+Comment for 6.3.7. $\mathbb{R}$이 아니라 일반적인 F 에서 가능, 하지만 $ch(F) = 0$ 이어야할
터 약화해도 될것같다, 귀찮다.